

Measuring the Effect of Shifted Chunk Boundaries on Evidence Retrieval

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Abstract— Fixed-size document chunking can divide supporting evidence differently depending on where chunk boundaries are placed, even when the source text and chunk width remain unchanged. Using 94 held-out Doc2Dial queries, this study examines the effect by shifting 200-word chunk boundaries across eight phases. The Evidence Fragmentation Score (EFS) measures the fraction of gold evidence not contained in the single chunk with the greatest evidence overlap. Retrieval is evaluated with BM25 and E5-base-v2 using a controlled ranking probe with fixed distractors and a full-index within-document experiment. When each query’s lowest- and highest-fragmentation conditions were compared, mean reciprocal rank decreased from 0.767 to 0.588 for BM25 and from 0.715 to 0.644 for E5. A document-cluster bootstrap kept the BM25 interval below zero, while the E5 interval widened to include zero. The full-index metrics changed in the same direction. In a matched four-phase ablation, adding 40 words of overlap reduced mean EFS from 0.051 to less than 0.001. Because EFS became constant across phases for 92 of 94 queries, the planned low-versus-high fragmentation retrieval comparison could no longer be evaluated for most of the held-out set. These results show that fixed-size boundary placement can produce measurable changes in evidence retrieval under otherwise controlled conditions, while the effect can still vary across individual queries.

Index Terms— document chunking, evidence retrieval, evidence fragmentation, information retrieval, retrieval-augmented generation

I. INTRODUCTION

Retrieval-augmented generation (RAG) systems depend on retrieving useful passages from external documents before generation [1]. Long documents are therefore often divided into smaller chunks that can be indexed and ranked separately. The way these chunks are formed matters because it determines which parts of the source text are available together inside a retrievable passage.

Fixed-size chunking is simple and inexpensive, but its boundaries do not necessarily follow the meaning or structure of a document. The same supporting evidence can fall completely inside one chunk under one boundary placement and be divided across neighboring chunks after the boundary pattern is shifted. In that case, the source document, query, and chunk width remain unchanged, but the evidence presented to the retriever is different.

Previous work has shown that document segmentation and chunk construction can affect retrieval and RAG performance [2], [3]. However, comparisons between different chunking methods can change several properties at once, such as boundary locations, chunk lengths, context, and the number of passages. This makes it difficult to separate the effect of boundary placement from the other differences between chunking methods.

This study examines whether shifting fixed-size document chunk boundaries changes the retrievability of gold supporting evidence when the document, query, chunk width, retriever, and retrieval settings remain unchanged. The hypothesis is that greater fragmentation of the gold evidence across chunk boundaries will be associated with worse ranking of the best available evidence-bearing passage.

The experiment uses Doc2Dial examples with exact annotated document-grounding spans as gold evidence. Each source document is divided into 200-word chunks under eight shifted boundary phases. EFS measures how much gold evidence is missing from the chunk that contains the largest share. Retrieval is evaluated with BM25 and E5-base-v2 using a controlled rank-

ing probe with fixed distractors and a full-index experiment. A 40-word overlap condition tests a simple way to reduce fragmentation.

Under the tested conditions, higher fragmentation was associated on average with poorer evidence retrieval for both retrievers. The decrease was larger for BM25 than for E5, and the full-index evaluation also showed lower retrieval performance at higher fragmentation. A 40-word overlap ablation sharply reduced fragmentation but left too little EFS variation for an aggregate retrieval-mitigation comparison. These results do not show that one chunking strategy is best in general, but they do show that fixed-size boundary placement can change retrieval even when the document and retrieval settings stay fixed.

II. RELATED WORK

Recent work has looked more directly at how document segmentation affects retrieval. Wang et al. describe document segmentation as an important but underexplored part of RAG and propose PIC, a method that uses document summaries to guide how related sentences are grouped into chunks [2]. Their experiments show that changing the segmentation method can affect both retrieval and question-answering performance. Qu et al. compare fixed-size and semantic chunking across document retrieval, evidence retrieval, and answer generation [3]. They find that semantic chunking can help in some settings, but the gains are inconsistent and often do not justify the additional computational cost. Their work also highlights a basic weakness of fixed-size chunking: a fixed boundary can separate related content that would ideally remain together.

Other work has focused on measuring chunk quality or preserving context across chunk boundaries. Zhao et al. introduce Boundary Clarity and Chunk Stickiness as direct measures of chunking quality and use them in the development of the MoC chunking framework [4]. Günther et al. approach the problem from the embedding side with late chunking [5]. Instead of independently embedding each chunk, late chunking first processes

a longer context and then forms chunk embeddings from the contextualized token representations. This is meant to reduce the loss of surrounding context that can happen when chunks are embedded separately.

These studies show that chunk construction can affect retrieval, but they generally compare different segmentation strategies, introduce new chunking methods, or change how chunk representations are produced. Unlike those comparisons, this study isolates boundary placement within fixed-size chunking and uses known gold-evidence locations to measure fragmentation and changes in retrieval rank. A separate overlap ablation tests whether a fixed overlap setting reduces evidence fragmentation.

III. METHOD

Table I summarizes the main experimental configuration.

Table I: Main experimental configuration.

Setting	Value
Dataset	Doc2Dial v1.0.1
Held-out set	94 queries, 46 documents
Chunk width	200 source words
Primary overlap	0 words
Boundary phases	0, 25, 50, 75, 100, 125, 150, 175
Retrieval scope	Known source document
Retrievers	BM25 and E5-base-v2
Primary outcome	Reciprocal Rank
Validation metrics	graded nDCG@5; oracle-normalized Evidence Coverage@5

A. Dataset Construction

Doc2Dial is the source dataset. It contains goal-oriented conversations grounded in documents, with dialogue turns linked to specific parts of the source text [6]. Its annotated grounding spans serve as the supporting evidence that a retriever should find.

The dialogue data were converted into retrieval examples using fixed rules. For each dialogue, only the first user turn and the immediately following agent turn were considered. An example was kept if the agent turn contained a `solution` or `precondition` grounding reference. Each reference was resolved to its exact source-document position, and the annotated text was checked against the original. Examples with multiple grounding spans were kept only when the spans formed one continuous evidence region. An unsuitable first exchange was not replaced with a later one.

Boundary-shift filtering required evidence shorter than the 200-word chunk size and positioned so that at least one tested phase could split it. Examples were removed when clipped chunks at the beginning or end of a document would interfere with the comparison. Evidence positions were converted from character offsets to source-word coordinates so that evidence and chunk boundaries used the same word definition. For both segmentation and evidence coordinates, a source word was a maximal non-whitespace span, equivalent to `\S+`.

The official training dialogues were used for development and implementation checks, while the official validation dialogues were kept for the final held-out experiment. After filtering, the

development set contained 472 examples from 111 documents. The held-out set contained 94 queries from 46 documents. In the held-out set, the gold evidence had a median length of 25 words and a maximum length of 58 words.

B. Shifted Boundary Segmentation

The experiment used fixed-size chunks of 200 source words and changed only the boundary positions. The main experiment used eight boundary phases: 0, 25, 50, 75, 100, 125, 150, and 175 words. This covers one full 200-word boundary cycle while keeping the chunk width unchanged.

At phase 0, the document was divided using the normal fixed-word chunking procedure. For a nonzero phase, the regular 200-word chunk pattern began at the selected offset. For example, at phase 50, the first 50 words formed a clipped prefix and the remaining text was divided into 200-word chunks beginning immediately after that prefix. The clipped prefix was kept so that shifting the boundary pattern did not drop any source words.

All chunk positions were mapped back to the coordinates of the original document. This allowed the gold evidence span and every phase’s chunks to use the same source-word positions. As a result, the document content and chunk width stayed fixed while only the placement of the boundaries changed.

C. Evidence Fragmentation Score

The *Evidence Fragmentation Score* (EFS) measures how strongly gold evidence is split across chunks. Let E be the gold evidence span and C the chunks from one boundary phase. EFS is defined as

$$\text{EFS}(E, C) = 1 - \max_{c \in C} \frac{|E \cap c|}{|E|} \quad (1)$$

The score is based on the chunk that contains the largest part of the evidence. If one chunk contains the full evidence span, then EFS is 0. A larger value means that no single chunk contains as much of the evidence, so the evidence is more fragmented across chunk boundaries.

EFS was computed from source-word positions rather than character counts. This kept the measurement consistent with the fixed-word chunking procedure. Across the held-out queries, every query had at least one phase with $\text{EFS} = 0$, and every query also had at least one phase with EFS greater than 0. The mean of each query’s maximum EFS was 0.303.

D. Retrieval Models

The study evaluates BM25 and E5-base-v2. BM25 is the sparse lexical retriever and ranks passages using term frequency and document-length normalization [7]. The implementation uses `rank_bm25.BM25Okapi`. Text is lowercased and tokenized with regular-expression word tokens matching `\w+`; this tokenizer is separate from the source-word definition used for segmentation and evidence coordinates. The fixed settings are $k_1 = 1.5$, $b = 0.75$, and $\epsilon = 0.25$.

Dense retrieval uses `intfloat/e5-base-v2`. E5 is a family of general-purpose text embedding models trained with contrastive learning for tasks such as retrieval [8]. Its retrieval setup uses separate `query:` and `passage:` prefixes. The

implementation adds the appropriate prefix, normalizes embeddings, and ranks passages by dot product between normalized vectors. Every input is checked against the 512-token limit so that passages are not silently truncated.

Retrieval lists used original passage order as a deterministic tie break for equal scores. For controlled-probe RR, an evidence passage tied with other candidates instead received the average tied rank before the reciprocal was taken, so deterministic tie order did not affect RR.

E. Controlled Ranking Probe

For each query and phase, the chunk containing the largest fraction of gold evidence was the *best available evidence passage*. If multiple chunks tied, the earlier chunk in document order was selected.

For each query, the distractor set consisted of all phase-0 chunks from the same source document with no gold-evidence overlap. This set was reused across all boundary phases, while the evidence passage was selected separately for each phase.

The phase-specific evidence passage was then ranked with the fixed distractors using BM25 and E5. This setup reduced candidate-set variation by holding the distractor texts and candidate count fixed while the shifted boundaries changed the evidence passage. For E5, the score of a fixed distractor depends only on the query and that passage, so the unchanged distractor passages were scored independently of the phase-specific evidence passage. For BM25, however, `BM25Okapi` was fit separately to each phase-specific candidate set. Changing the evidence passage could therefore also change candidate-set statistics such as average passage length and query-term document frequencies, which can affect the scores of otherwise unchanged distractors. The controlled probe therefore fixes the distractor texts rather than every component of the BM25 scores.

F. Full-Index Validation

A full-index experiment checked whether the controlled-probe pattern remained when all phase-specific chunks were used. For each query and boundary phase, all chunks from the known source document were ranked with BM25 and E5.

Because a phase can split the same evidence across several chunks, graded relevance reflects how much gold evidence each chunk contains. `nDCG@5` measures whether chunks with more evidence rank near the top. Oracle-normalized Evidence Coverage@5 measures top-five gold-evidence coverage relative to the best coverage possible for that phase.

Unlike the controlled probe, the candidate set changed with each boundary phase because the entire shifted chunk collection was ranked.

G. Evaluation and Statistical Analysis

The main outcome for the controlled ranking probe was reciprocal rank (RR) of the best available evidence passage. For a passage at rank r , RR is $1/r$. A passage ranked first therefore has an RR of 1, while lower-ranked passages receive smaller values. If the evidence passage tied with other candidates, its rank was the average of the tied positions. For each query

and retriever, the lowest-EFS phases were compared with the highest-EFS phases. RR values were averaged when several phases shared the same minimum or maximum EFS. The paired difference is high-EFS RR minus low-EFS RR, so a negative value means worse retrieval at higher fragmentation.

A query-level bootstrap estimated uncertainty. Queries were resampled with replacement 10,000 times to form a 95% percentile confidence interval for the mean paired RR difference. The query was used as the statistical unit so that the eight boundary phases from the same query were not treated as independent examples.

Because multiple held-out queries can come from one source document, a document-cluster bootstrap provided a sensitivity analysis. The 46 source documents were resampled with replacement, including all query-level paired RR differences for each sampled document. The statistic remained the mean query-level RR difference. This sensitivity analysis also used 10,000 resamples and a 95% percentile interval.

Both bootstrap procedures used random seed 0.

A secondary analysis calculated the within-query Spearman rank correlation between EFS and RR across the eight phases. This measured whether RR tended to decrease as EFS increased within the same query. The correlation was left undefined when either EFS or RR was constant across all phases for a query.

The full-index experiment used graded `nDCG@5` and oracle-normalized Evidence Coverage@5. Because a phase can split the same evidence across several chunks, graded relevance was defined as the fraction of gold evidence contained in each chunk. `nDCG@5` follows the normalized discounted cumulative gain framework of Järvelin and Kekäläinen [9]. In the implementation, a chunk with graded relevance rel at rank r contributed $(2^{rel} - 1) / \log_2(r + 1)$, and `DCG@5` was normalized by the ideal ordering for that phase. Evidence Coverage@5 counted the unique gold-evidence source words covered by the union of the evidence-overlap intervals from the top five retrieved chunks, so overlapping chunks did not double-count evidence. This value was normalized by the maximum unique gold-evidence coverage from any set of up to five chunks in the same phase. Low- and high-EFS conditions used the same per-query extreme-phase rule as RR. Full-index results are paired means; bootstrap confidence intervals were not computed for these measures.

H. Overlap Ablation

One overlap ablation tested whether overlap between neighboring chunks reduced evidence fragmentation. The chunk width remained fixed at 200 words, while overlap was increased from 0 to 40 words. This changed the chunk stride from 200 words to 160 words.

Because the stride changed, each condition used four evenly spaced phase positions over one complete boundary cycle. The zero-overlap baseline used phases 0, 50, 100, and 150, while the 40-word-overlap condition used phases 0, 40, 80, and 120. These phase sets use the same quarter-stride positions within their respective cycles, making the two conditions more directly comparable.

The two conditions were compared mainly by EFS. For each query, mean EFS was calculated across the four matched phase positions, along with its decrease after adding overlap. A query-

level bootstrap with 10,000 resamples estimated a 95% confidence interval for the mean reduction. The overlap condition was also checked for enough within-query EFS variation to support the planned low-versus-high retrieval comparison, which was valid only when minimum and maximum EFS differed.

IV. RESULTS

A. Controlled Retrieval

In the controlled ranking probe, reciprocal rank was lower at the highest-fragmentation condition than at the lowest-fragmentation condition for both retrievers. For BM25, mean RR decreased from 0.767 to 0.588, giving a mean paired difference of -0.179 with a 95% query-bootstrap confidence interval of $[-0.238, -0.120]$. For E5, mean RR decreased from 0.715 to 0.644, with a smaller mean difference of -0.071 and a 95% query-bootstrap confidence interval of $[-0.131, -0.010]$.

Figure 1 shows the phase-level RR values across the observed EFS values for both retrievers.

Table II: Controlled retrieval results at the lowest and highest EFS conditions. The difference is high-EFS RR minus low-EFS RR; CIs use the primary query bootstrap.

Retriever	Low EFS	High EFS	Δ RR	95% CI
BM25	0.767	0.588	-0.179	$[-0.238, -0.120]$
E5	0.715	0.644	-0.071	$[-0.131, -0.010]$

In the document-cluster sensitivity analysis, the 95% interval for BM25 was $[-0.249, -0.095]$, which remained entirely below zero. For E5, the interval widened to $[-0.182, 0.037]$ and included zero. The mean differences did not change because this analysis changed only how the data were resampled.

The result was not the same for every query. At higher fragmentation, BM25 RR decreased for 49 queries, stayed unchanged for 32, and increased for 13. E5 RR decreased for 43 queries, stayed unchanged for 22, and increased for 29.

The within-query correlations were also more negative for BM25. Among queries where the correlation was defined, the mean Spearman correlation between EFS and RR was -0.518 for BM25 and -0.128 for E5. The correlation was undefined for 29 BM25 queries and 21 E5 queries because either EFS or RR was constant across phases.

The query-level results also showed different kinds of behavior. For a query about New York vehicle inspection rules, EFS increased from 0 to 0.429, while RR decreased by 0.750 for BM25 and 0.655 for E5. In contrast, a Premium Waiver query reached an EFS of 0.465 while both retrievers still ranked the evidence passage first. A query about obtaining a New York inspection extension showed disagreement between the retrievers: EFS increased from 0 to 0.167, BM25 RR decreased by 0.667, and E5 RR increased by 0.357. These examples match the aggregate result that fragmentation was associated with poorer retrieval on average, but not for every query or retriever.

B. Full-Index Results

The full-index experiment also showed lower retrieval quality at higher fragmentation. For BM25, mean nDCG@5 decreased

from 0.813 at low fragmentation to 0.708 at high fragmentation, a difference of -0.104 . Oracle-normalized Evidence Coverage@5 decreased from 0.982 to 0.922, a difference of -0.060 .

E5 showed smaller decreases. Mean nDCG@5 went from 0.768 to 0.717, giving a difference of -0.051 . Oracle-normalized Evidence Coverage@5 decreased from 0.964 to 0.940, a difference of -0.024 .

Table III summarizes these paired descriptive means.

Table III: Full-index validation results. Differences are high-EFS minus low-EFS values. Oracle-norm. Coverage@5 denotes oracle-normalized Evidence Coverage@5.

Retriever	Metric	Low	High	Δ
BM25	nDCG@5	0.813	0.708	-0.104
BM25	Oracle-norm. Coverage@5	0.982	0.922	-0.060
E5	nDCG@5	0.768	0.717	-0.051
E5	Oracle-norm. Coverage@5	0.964	0.940	-0.024

C. Overlap Ablation Results

Adding 40 words of overlap reduced evidence fragmentation in the matched four-phase comparison. Mean EFS decreased from 0.051 in the zero-overlap condition to less than 0.001 with overlap. The mean reduction was 0.051, with a 95% query-bootstrap confidence interval of $[0.042, 0.059]$. Mean EFS decreased for 67 queries, stayed unchanged for 27, and did not increase for any query.

Overlap removed almost all within-query EFS variation in the tested cycle. EFS was constant across the four overlap phases for 92 of 94 queries, leaving only two queries with distinct low- and high-fragmentation conditions. Because only two queries still had different low- and high-EFS conditions, the planned retrieval comparison could not be evaluated across the held-out set. An aggregate overlap RR or full-index mitigation effect is therefore not reported.

V. DISCUSSION

A. Main Finding

The main result supports the hypothesis under the conditions tested in this study. The best available evidence passage had a lower mean reciprocal rank at the highest fragmentation than at the lowest fragmentation. The controlled probe held distractor texts and candidate count fixed within each query while shifted boundaries changed the phase-specific evidence passage. As described in the method, BM25 was refit for each phase-specific candidate set, so its result should be interpreted as fixing distractor texts rather than every ranking-score component. The mean decrease was larger for BM25 than for E5.

The mixed query-level results suggest that EFS is related to retrieval difficulty on average, but it does not fully determine the ranking of an evidence passage. Query wording, the terms contained in each chunk, and the surrounding context can still affect the ranking.

One possible explanation for the larger BM25 decrease is that fragmentation can separate exact query-matching terms or supporting terms across neighboring passages, which may affect a lexical retriever more directly. A dense retriever may still represent some semantic similarity when only part of the

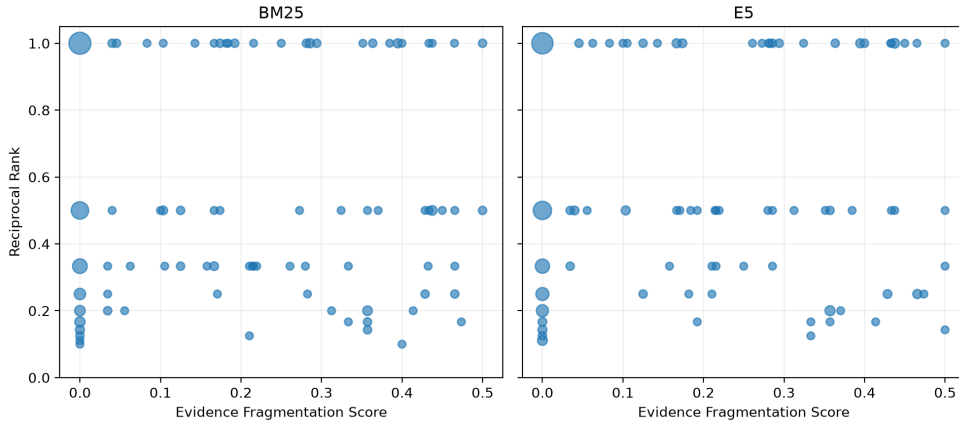


Figure 1: Reciprocal rank across observed Evidence Fragmentation Score values in the controlled probe for BM25 and E5. Marker size represents repeated observations at the same EFS and RR value.

evidence remains in a passage. No separate analysis of lexical-term placement or embedding behavior was run, so this remains a possible explanation rather than a tested mechanism. The results do not show that dense retrieval is generally resistant to boundary fragmentation; they only show a smaller average decrease for E5 under the conditions tested here.

B. Full-Index Validation and Overlap

The full-index experiment supports the controlled result under a broader within-document ranking setup. The controlled probe intentionally used fixed distractors, which makes the comparison easier to interpret, although the setup is less similar to a normal retrieval setting. Ranking all phase-specific chunks from the source document instead provided a check that the observed pattern was not limited to the fixed-distractor setup.

Because 40 words of overlap nearly eliminated within-query EFS variation, the ablation does not establish whether overlap reduces retrieval sensitivity to boundary placement.

The result should also not be interpreted as showing that 40 words is an optimal amount of overlap. The ablation evaluated a single overlap setting, and other overlap amounts could produce different fragmentation patterns. The held-out evidence spans were relatively short, with a median length of 25 words and a maximum length of 58 words. The added storage, indexing, and retrieval costs of producing more overlapping chunks were not measured.

C. Limitations

The study has several limitations. First, the experiment used only Doc2Dial. The dataset provides exact grounding spans, which made the boundary-shift study possible, but its documents and goal-oriented dialogue queries do not represent every type of RAG application. The filtering procedure also produced a specific subset of Doc2Dial. The filtered set kept examples with one continuous evidence region, evidence shorter than the chunk width, and evidence that could be fragmented by the tested phases. These findings apply most directly to examples with these properties rather than to the full dataset. The held-out experiment also contained only 94 queries from 46 documents, so the evaluation set is fairly small. The reported query-level

bootstrap intervals do not model possible dependence among multiple queries from the same source document. A document-cluster bootstrap sensitivity analysis evaluated this dependence for the primary RR result, but not for the overlap EFS-reduction interval. The sensitivity analysis produced a wider interval for E5 that included zero. This means the strength of the E5 result depends on whether queries are resampled individually or by source document.

Second, retrieval was restricted to the known source document. This kept the focus on chunk boundaries without adding document-selection errors, but it is simpler than corpus-scale retrieval. The controlled ranking probe is even more restricted because its distractors were fixed phase-0 passages from the same document. The full-index experiment reduces the fixed-distractor limitation by ranking every phase-specific chunk from the document, but it still does not test retrieval across a large collection of unrelated documents.

Third, the experiment used one primary chunk width, one overlap amount, and two retrievers. A 200-word chunk width was kept fixed to focus the experiment on boundary position, but other chunk widths could produce different fragmentation patterns. The overlap ablation tested only 40 words of overlap. Likewise, BM25 and E5-base-v2 provide useful sparse and dense retrieval settings, but the results do not establish the same behavior for every lexical, dense, hybrid, or reranking system.

EFS also describes only one aspect of chunk quality. It measures the fraction of gold evidence contained in the single chunk with the greatest evidence overlap. It does not directly measure semantic coherence, whether the missing part of an evidence span is essential, or whether related context outside the annotated span helps retrieval. Two passages with the same EFS can therefore still differ in how useful they are to a retriever.

Finally, this study evaluates retrieval rather than complete RAG answer quality. A lower evidence rank can make useful context harder to retrieve, but no language model was run to test whether the ranking changes altered final answers. The full-index results were also reported as paired descriptive means without bootstrap confidence intervals. These choices keep the conclusions focused on the retrieval stage.

VI. CONCLUSION

This study examined the effect of fixed-size chunk boundary placement on evidence retrieval by shifting 200-word boundaries while keeping the document, query, chunk width, retriever, and retrieval settings fixed. Using EFS to measure evidence fragmentation, the study found lower mean retrieval performance at higher fragmentation for both BM25 and E5-base-v2. The full-index evaluation supported this overall pattern. A secondary overlap ablation sharply reduced fragmentation but did not support an aggregate retrieval-mitigation estimate because too little within-query EFS variation remained.

These results show that chunk boundary placement can matter for evidence retrieval. At the same time, the effect varied across queries, and the study does not establish a universal chunking rule or a downstream effect on generated answers. Within the tested Doc2Dial setting, the results suggest that boundary placement and evidence fragmentation are worth measuring directly when evaluating document chunking for retrieval.

CODE AVAILABILITY

The code and materials needed to reproduce the experiments are available at: <https://github.com/kkorabal/chunk-boundary-retrieval>

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